

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

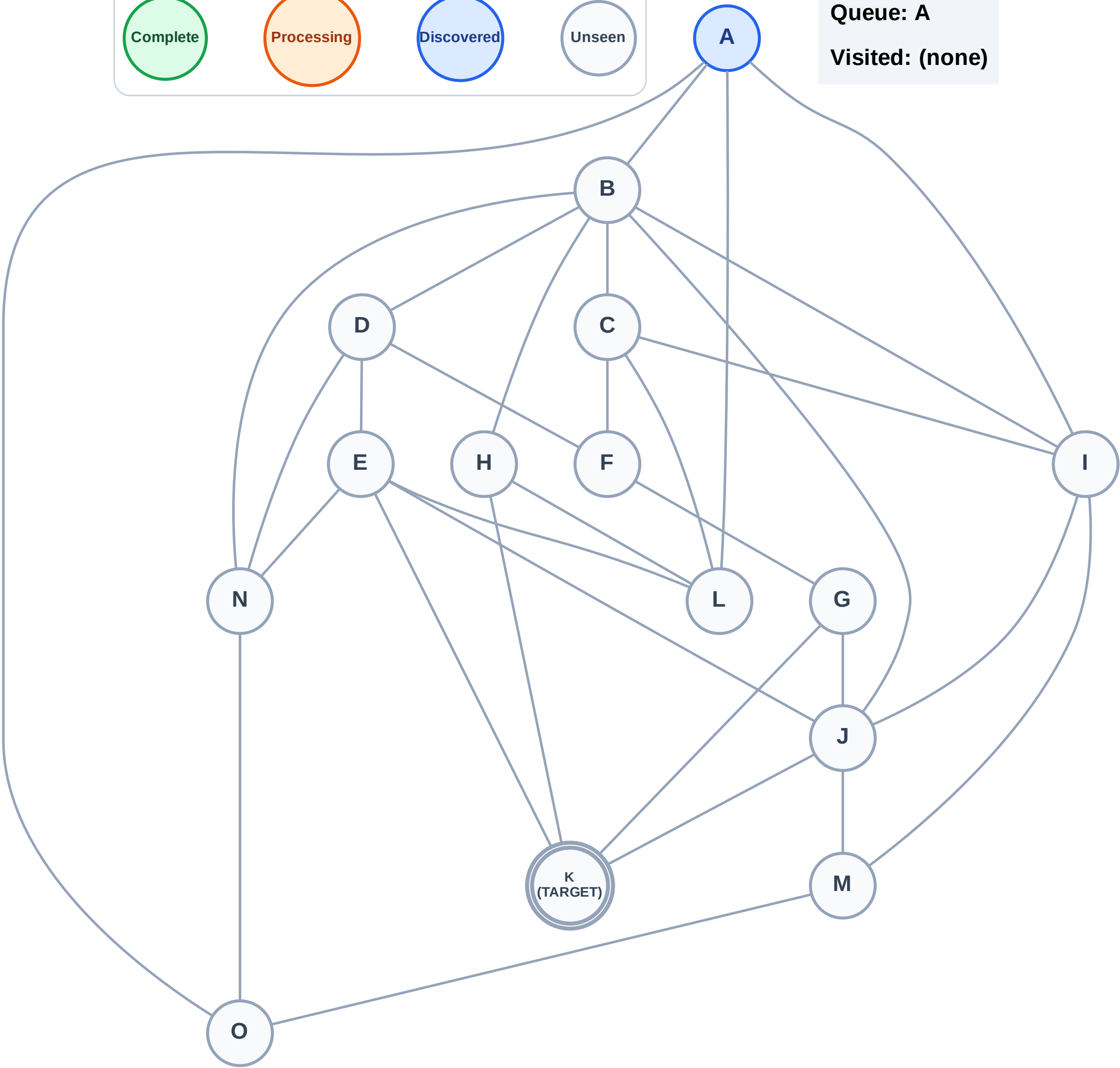
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

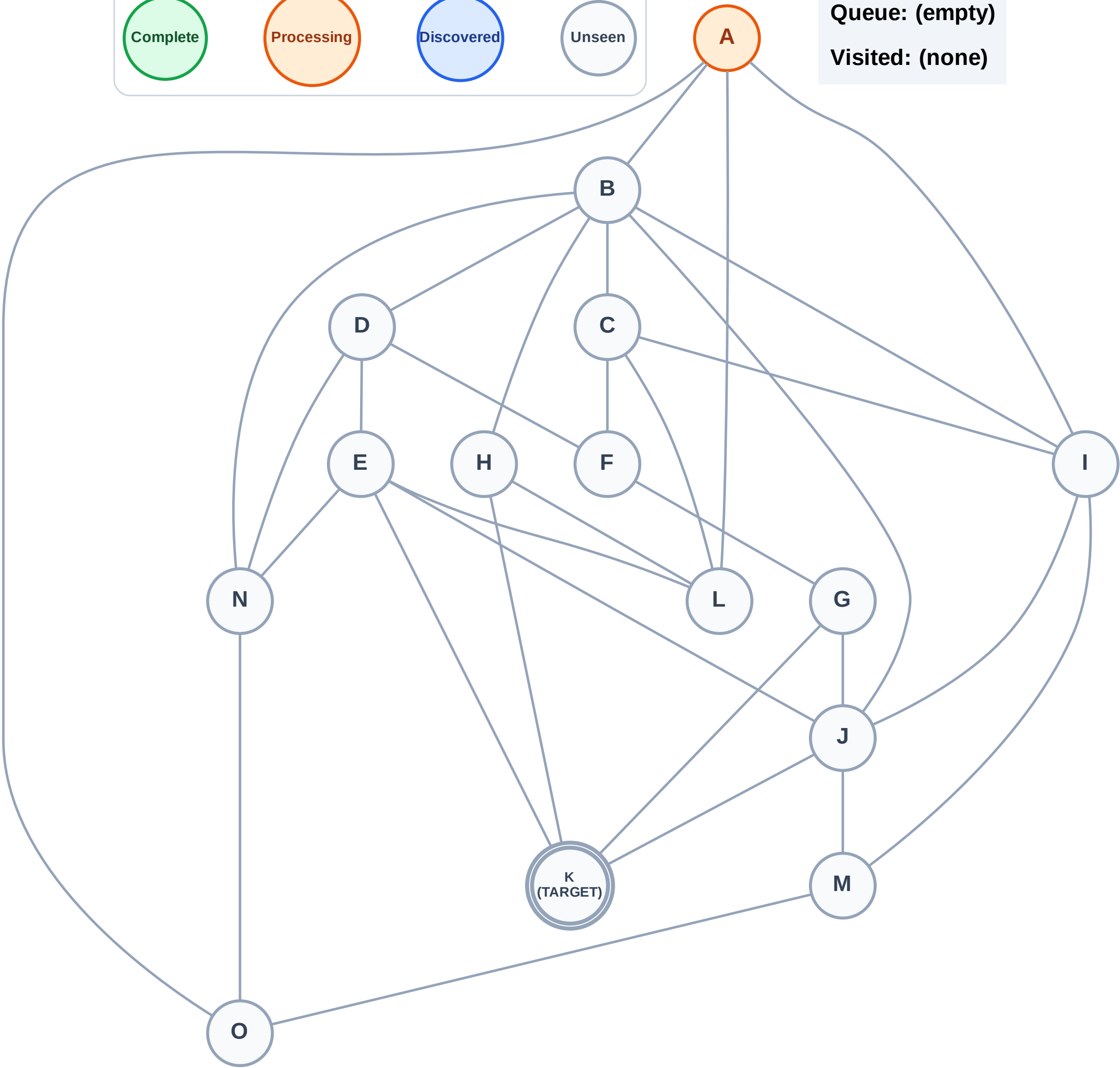
Complete

Processing

Discovered

Unseen

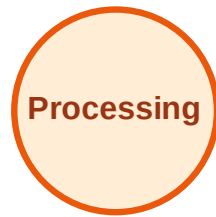
Queue: (empty)
Visited: (none)



BFS Traversal

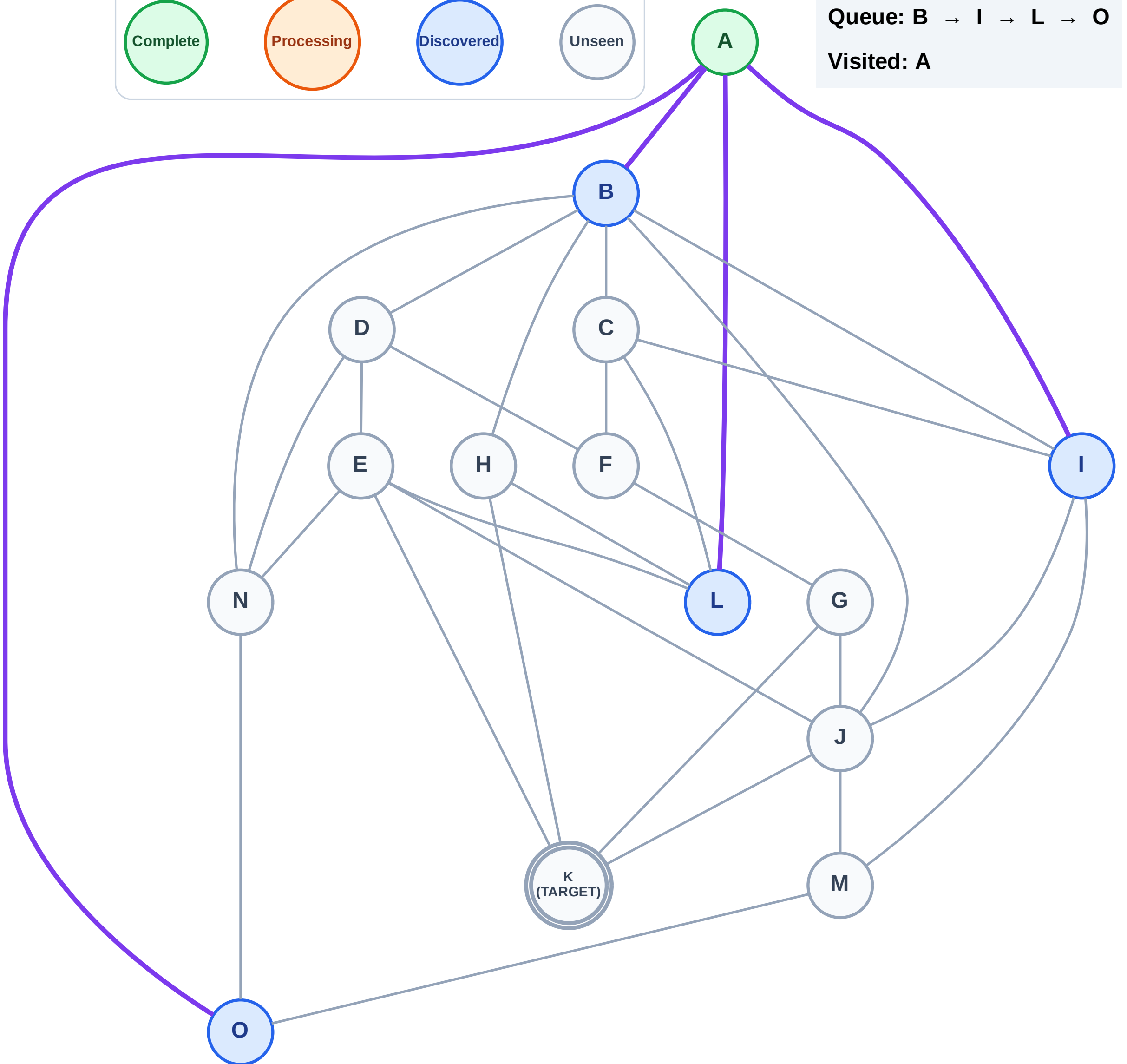
Step 3: Discover B, I, L, O from A

Legend



Queue: B → I → L → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

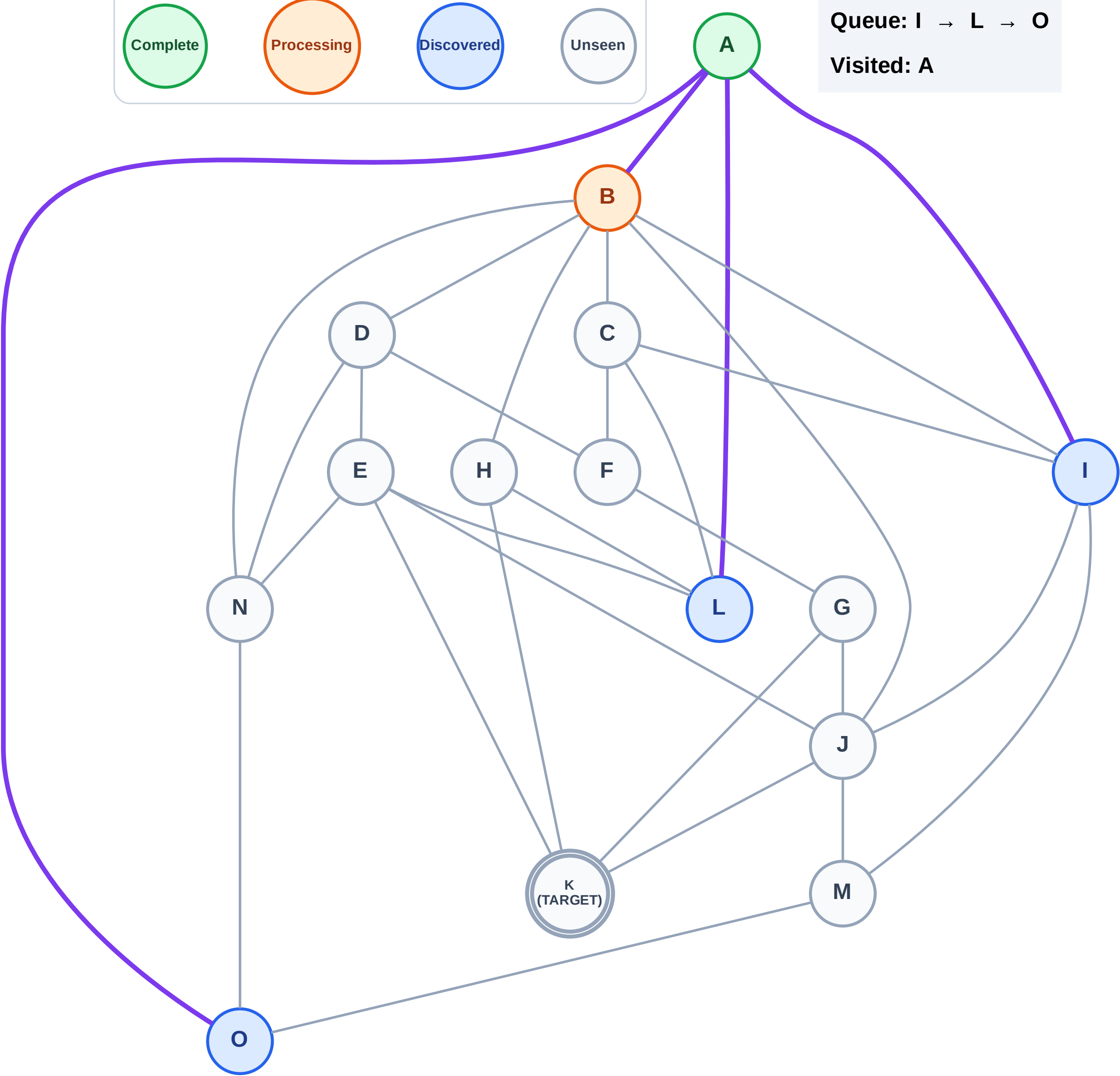
Processing

Discovered

Unseen

Queue: I → L → O

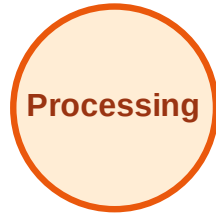
Visited: A



BFS Traversal

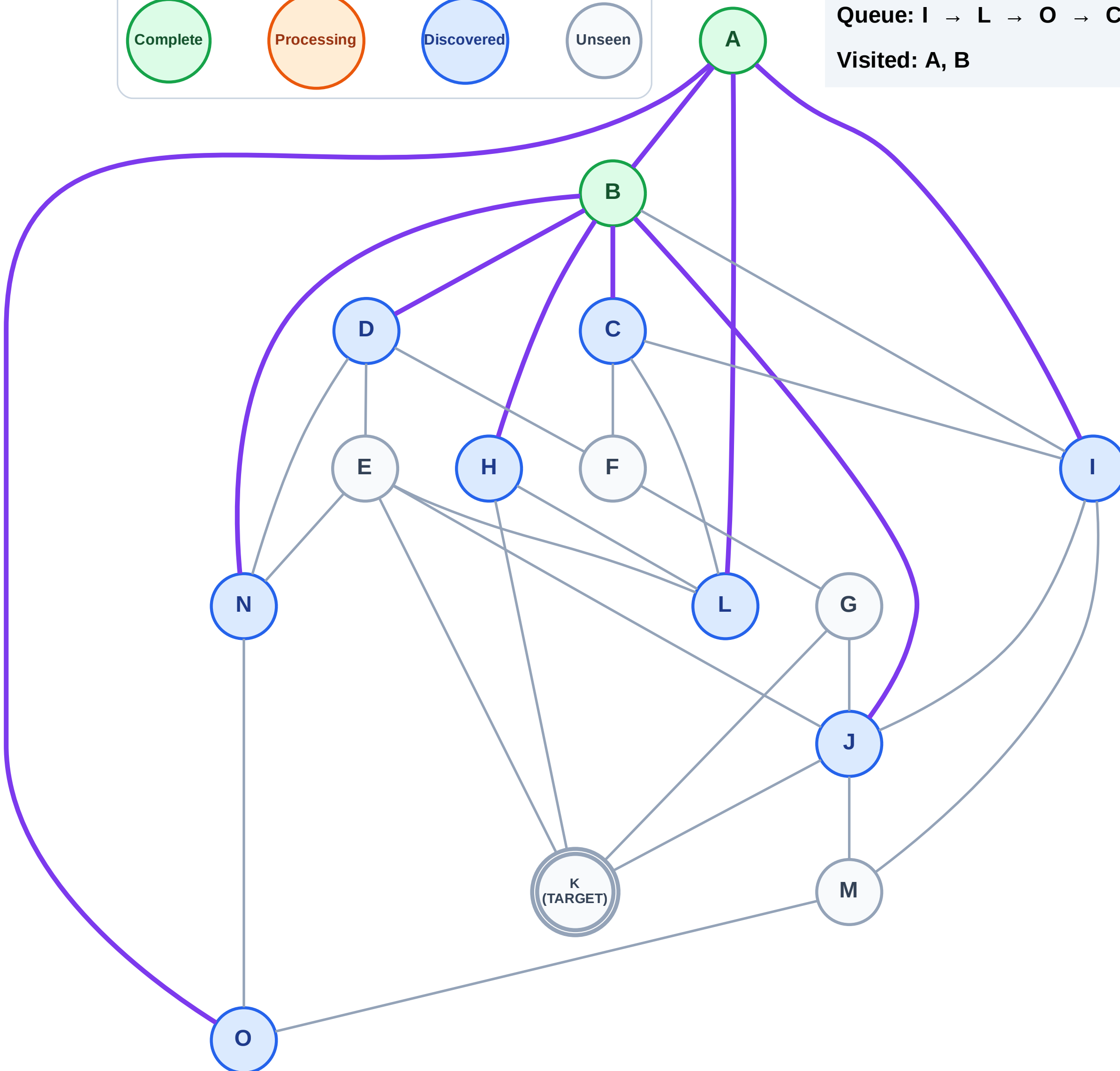
Step 5: Discover C, D, H, J, N from B

Legend



Queue: I → L → O → C → D → H → J → N

Visited: A, B



BFS Traversal

Step 6: Process node I

Legend

Complete

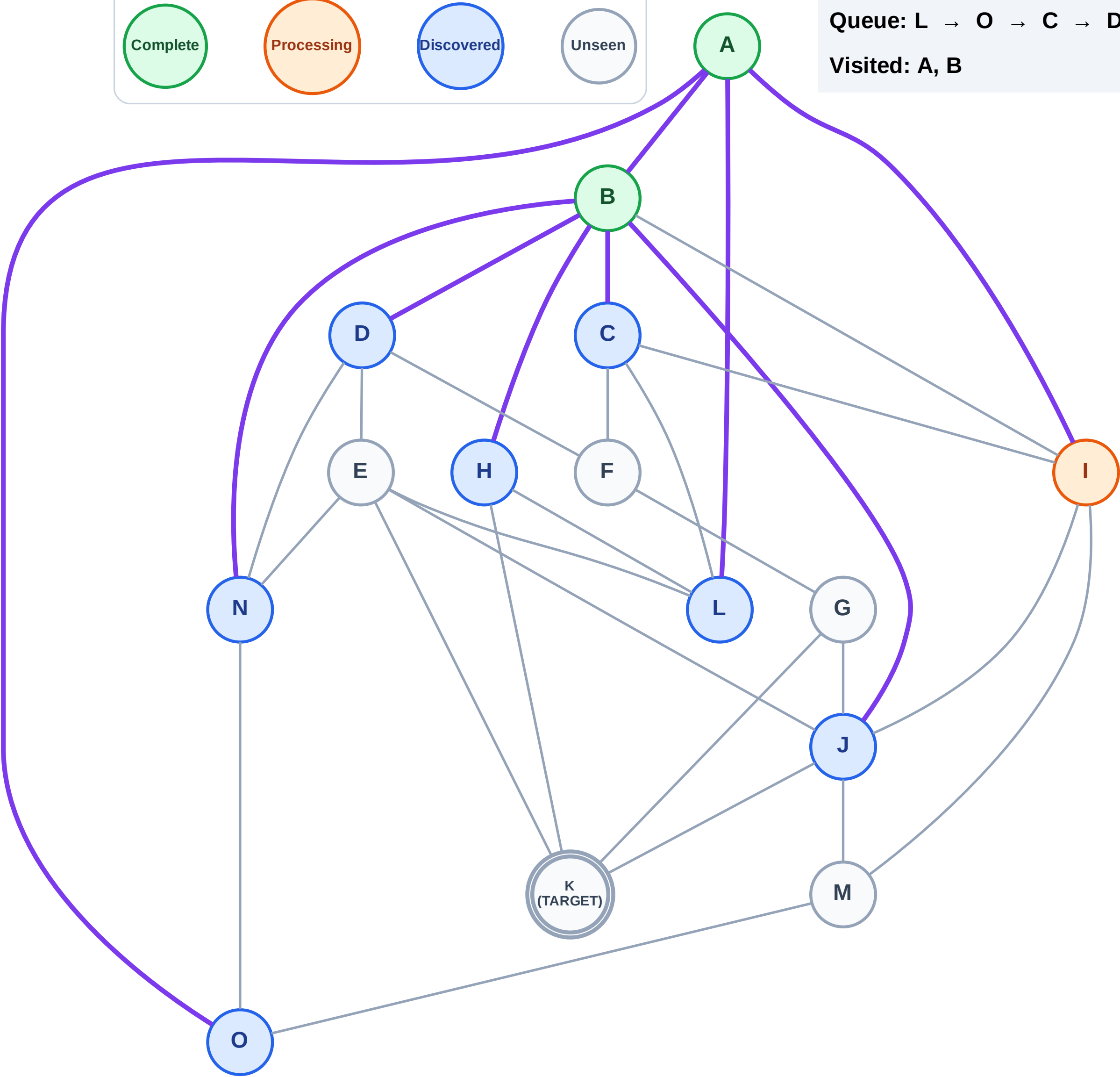
Processing

Discovered

Unseen

Queue: L → O → C → D → H → J → N

Visited: A, B



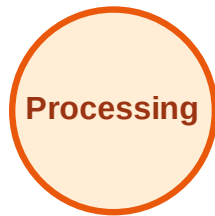
BFS Traversal

Step 7: Discover M from I

Legend



Complete



Processing



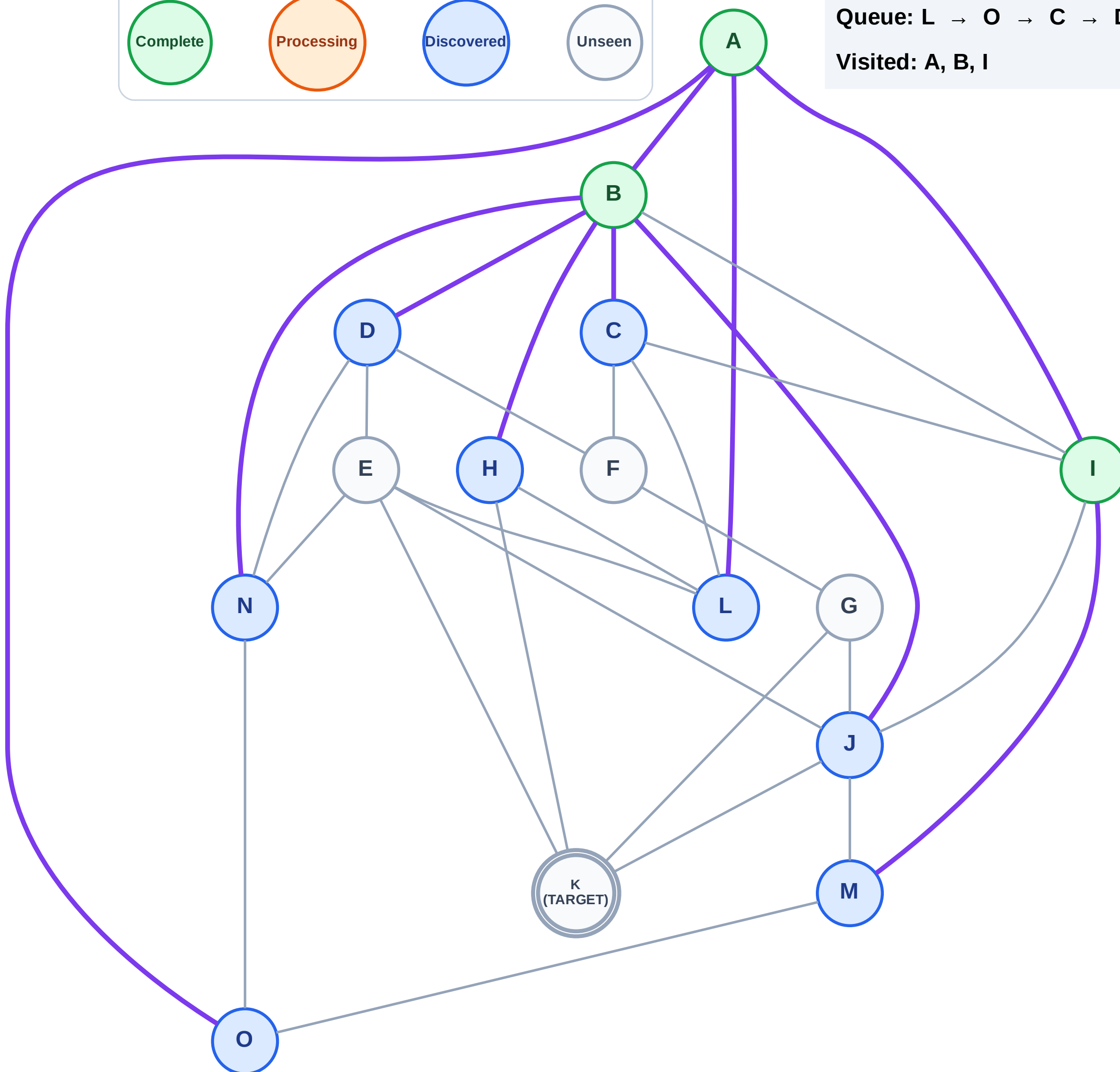
Discovered



Unseen

Queue: L → O → C → D → H → J → N → M

Visited: A, B, I



BFS Traversal

Step 8: Process node L

Legend

Complete

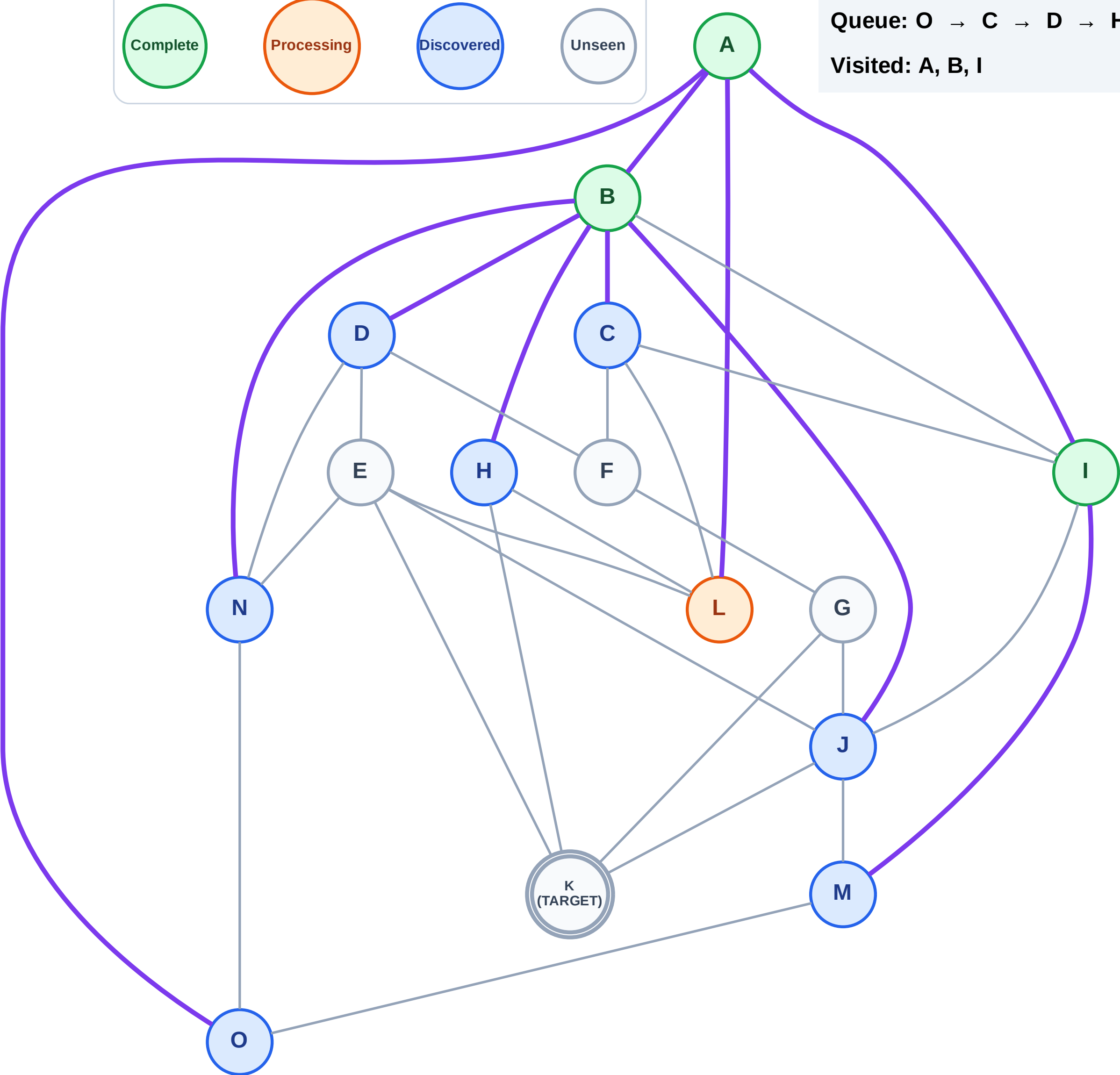
Processing

Discovered

Unseen

Queue: O → C → D → H → J → N → M

Visited: A, B, I



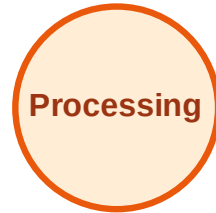
BFS Traversal

Step 9: Discover E from L

Legend



Complete



Processing



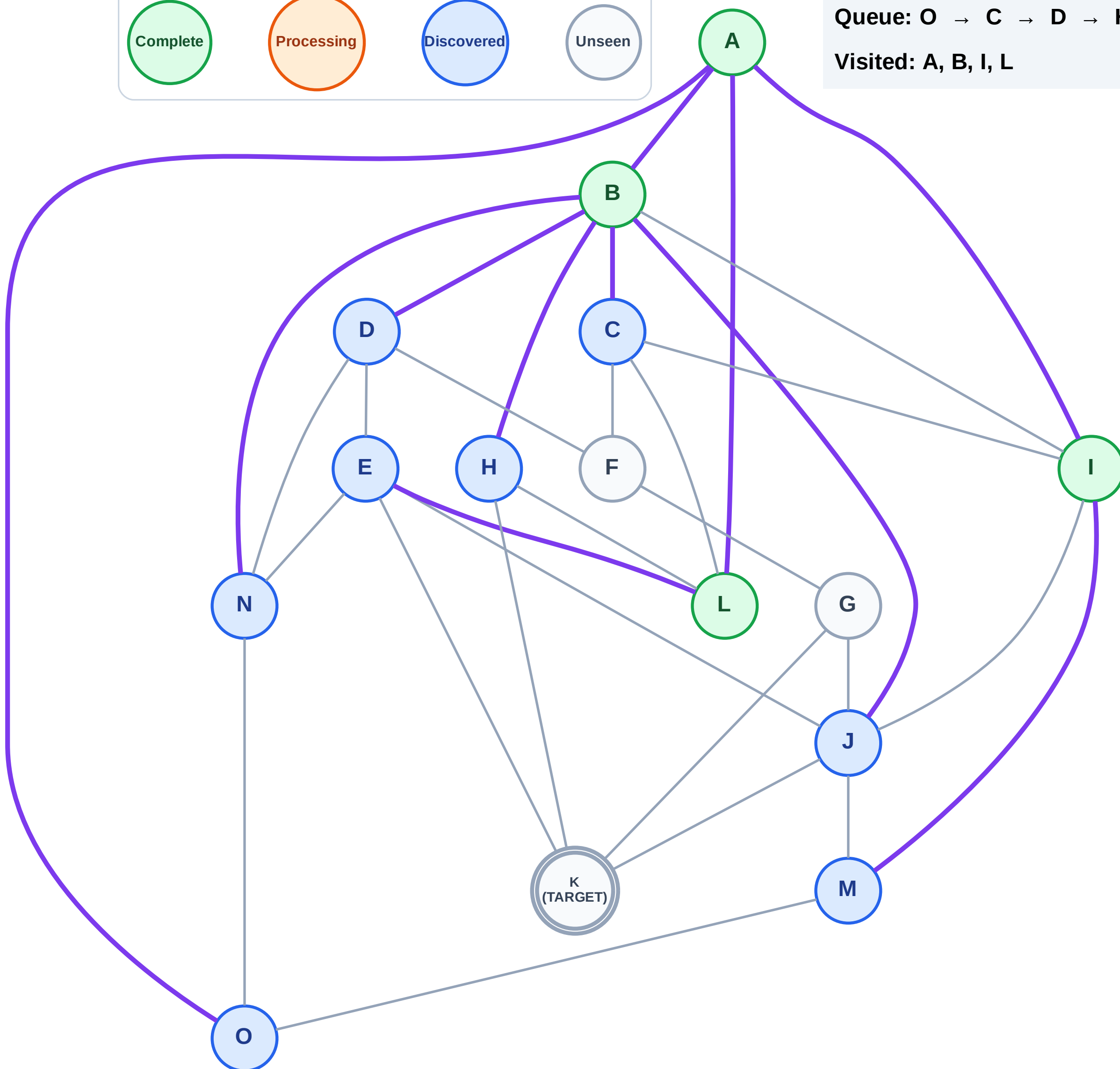
Discovered



Unseen

Queue: O → C → D → H → J → N → M → E

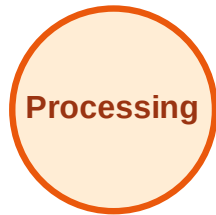
Visited: A, B, I, L



BFS Traversal

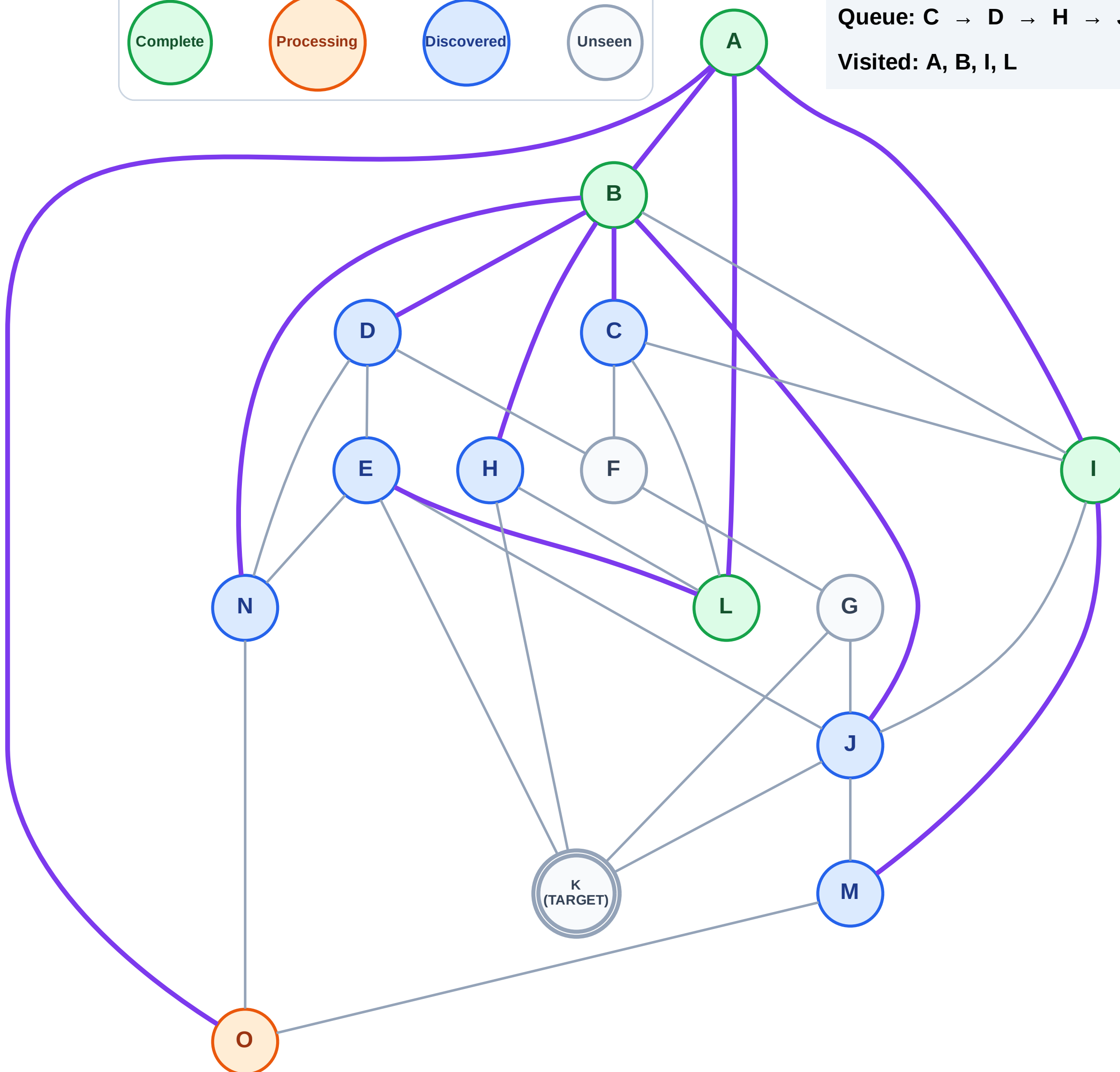
Step 10: Process node O

Legend



Queue: C → D → H → J → N → M → E

Visited: A, B, I, L



BFS Traversal

Step 11: No new neighbors from O

Legend

Complete

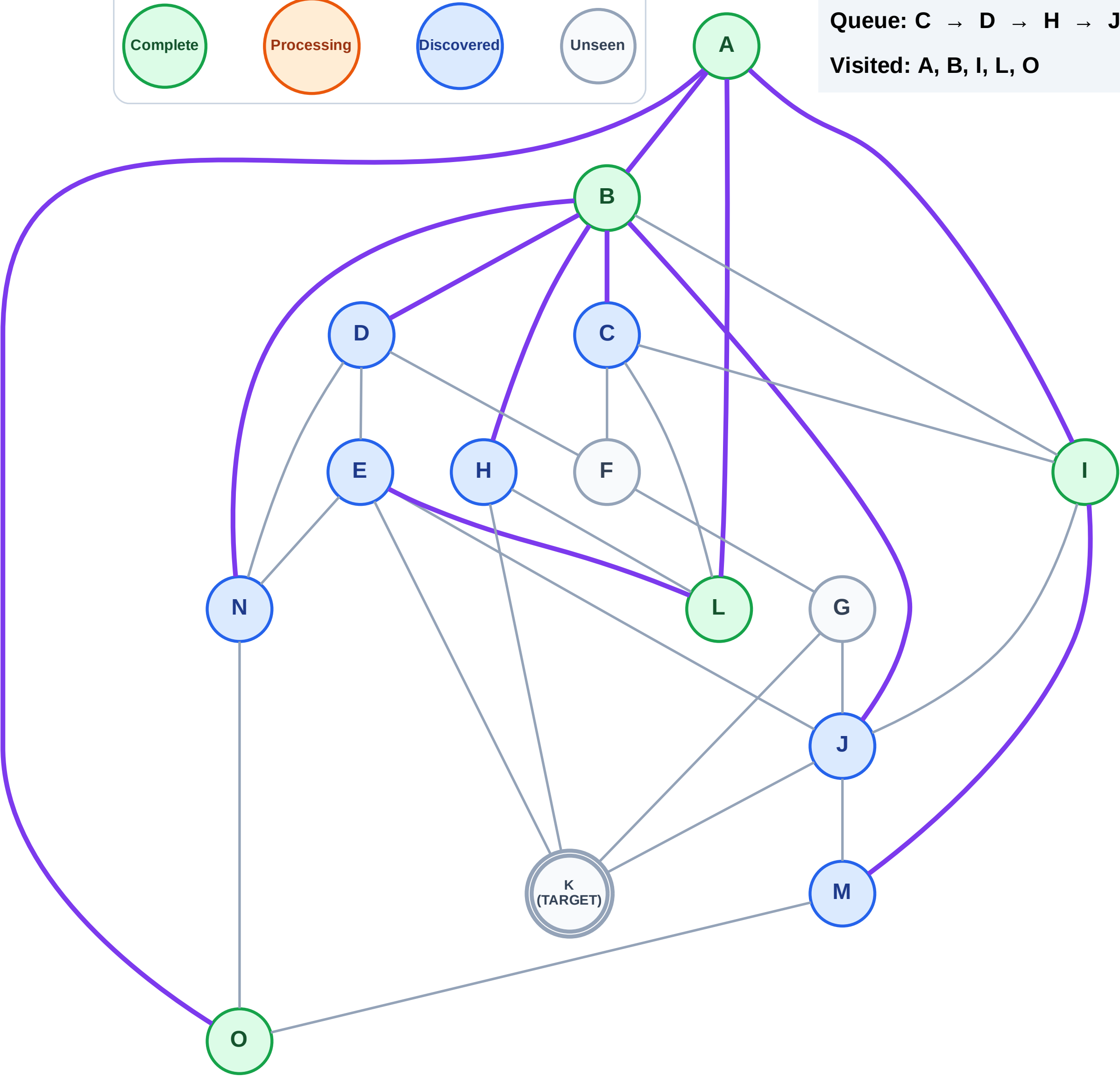
Processing

Discovered

Unseen

Queue: C → D → H → J → N → M → E

Visited: A, B, I, L, O



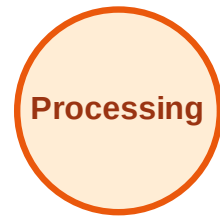
BFS Traversal

Step 12: Process node C

Legend



Complete



Processing



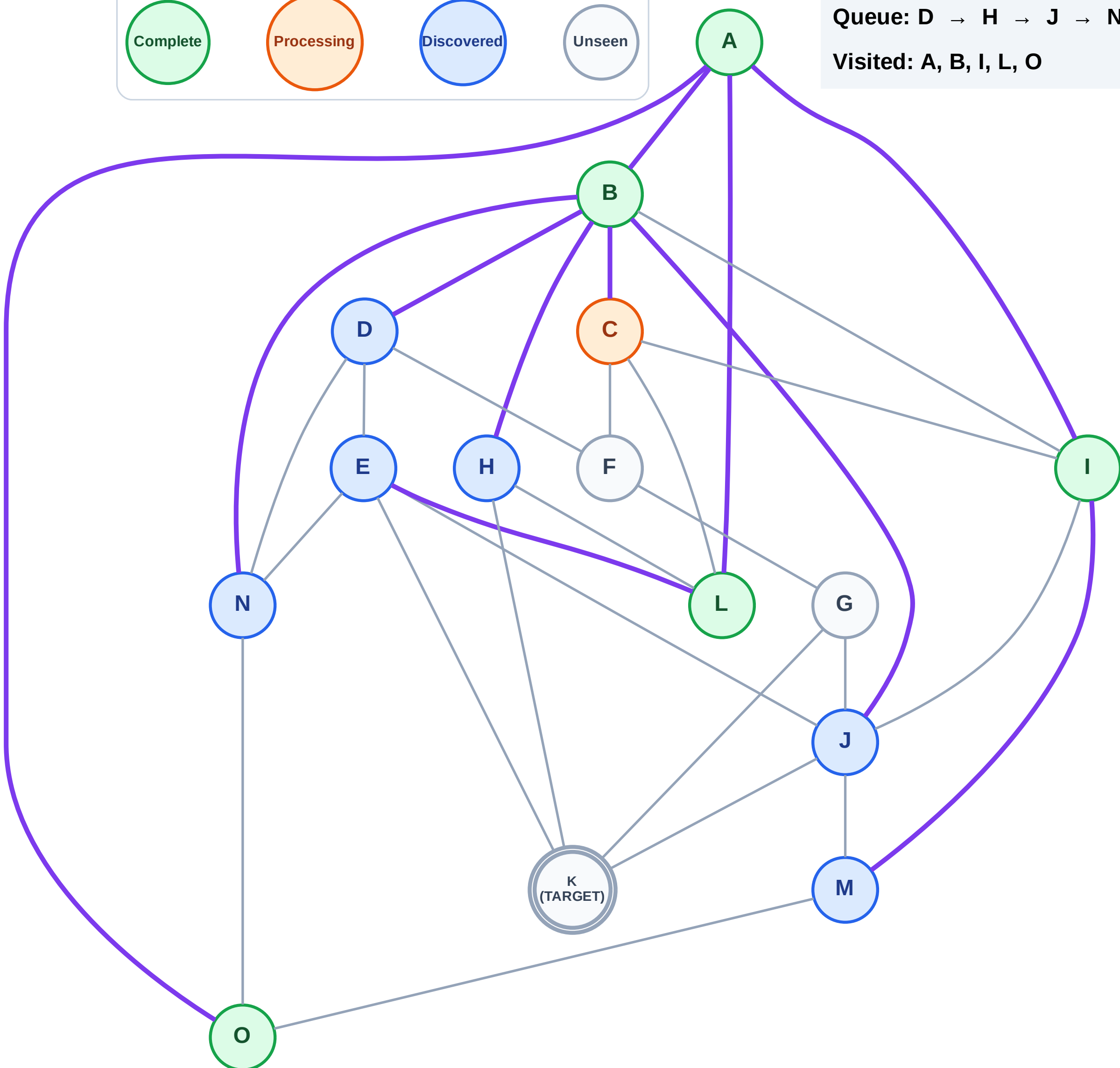
Discovered



Unseen

Queue: D → H → J → N → M → E

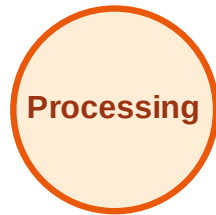
Visited: A, B, I, L, O



BFS Traversal

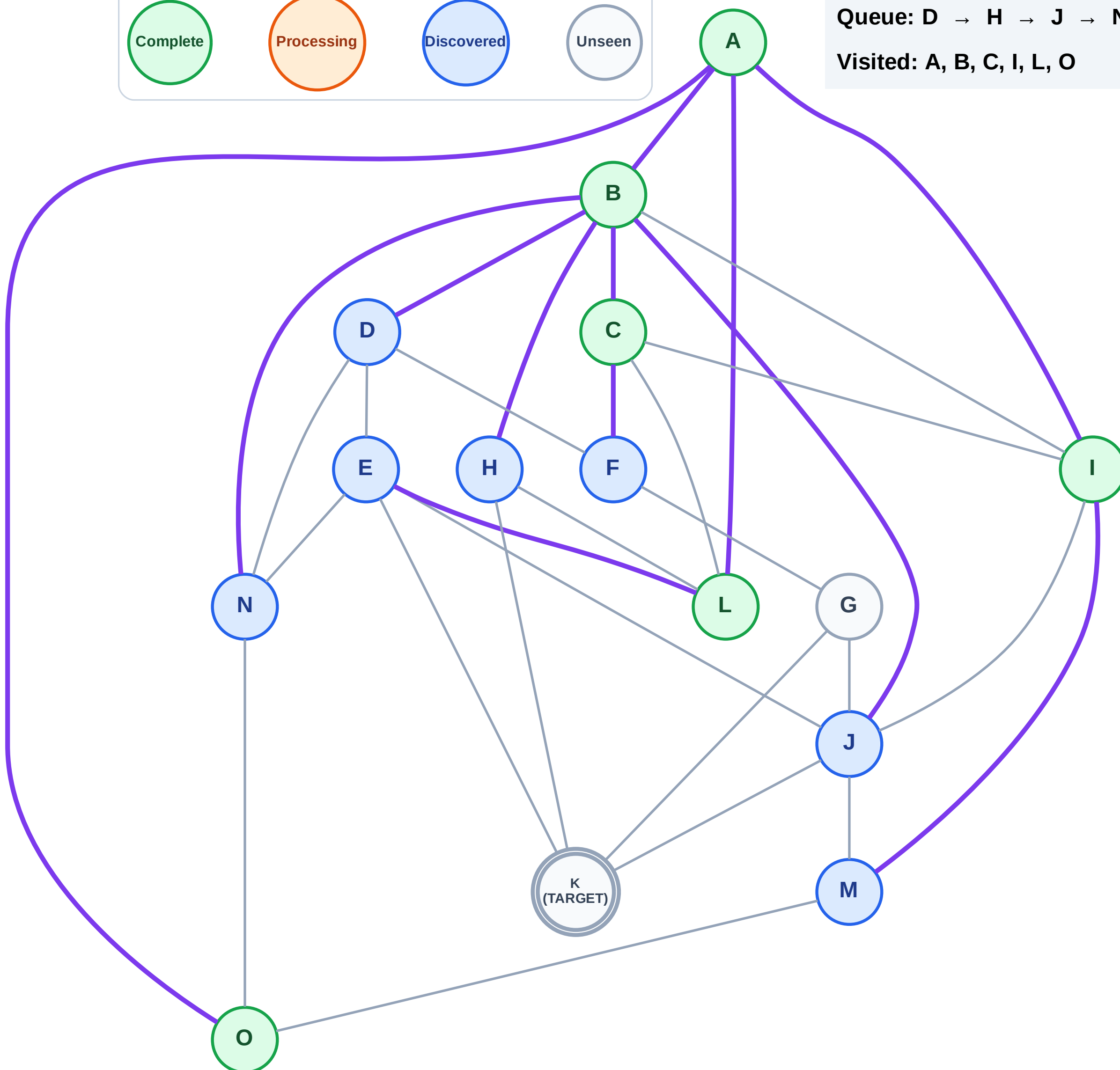
Step 13: Discover F from C

Legend



Queue: D → H → J → N → M → E → F

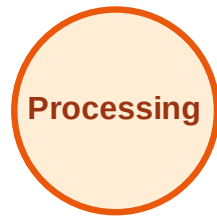
Visited: A, B, C, I, L, O



BFS Traversal

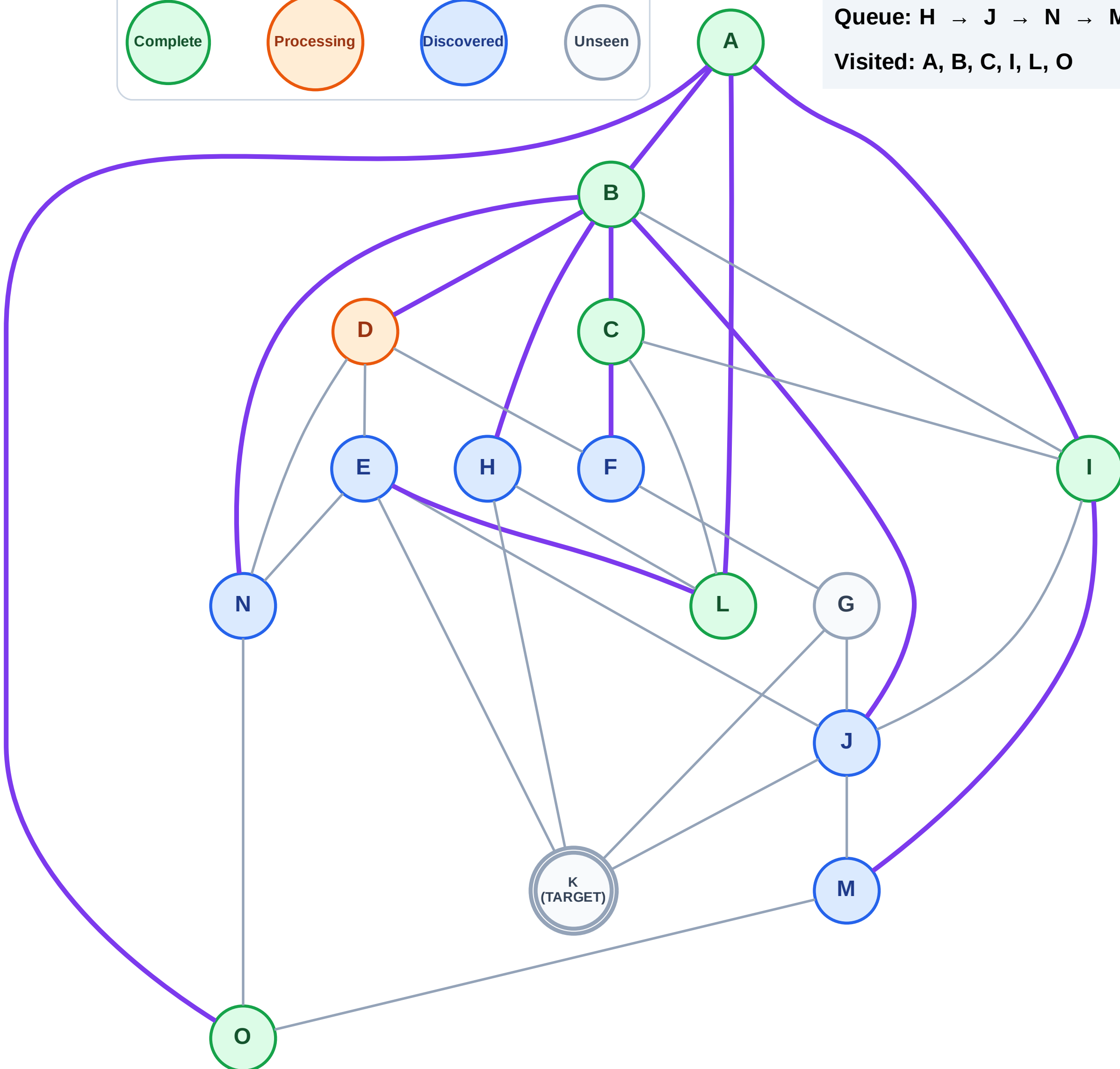
Step 14: Process node D

Legend



Queue: H → J → N → M → E → F

Visited: A, B, C, I, L, O



BFS Traversal

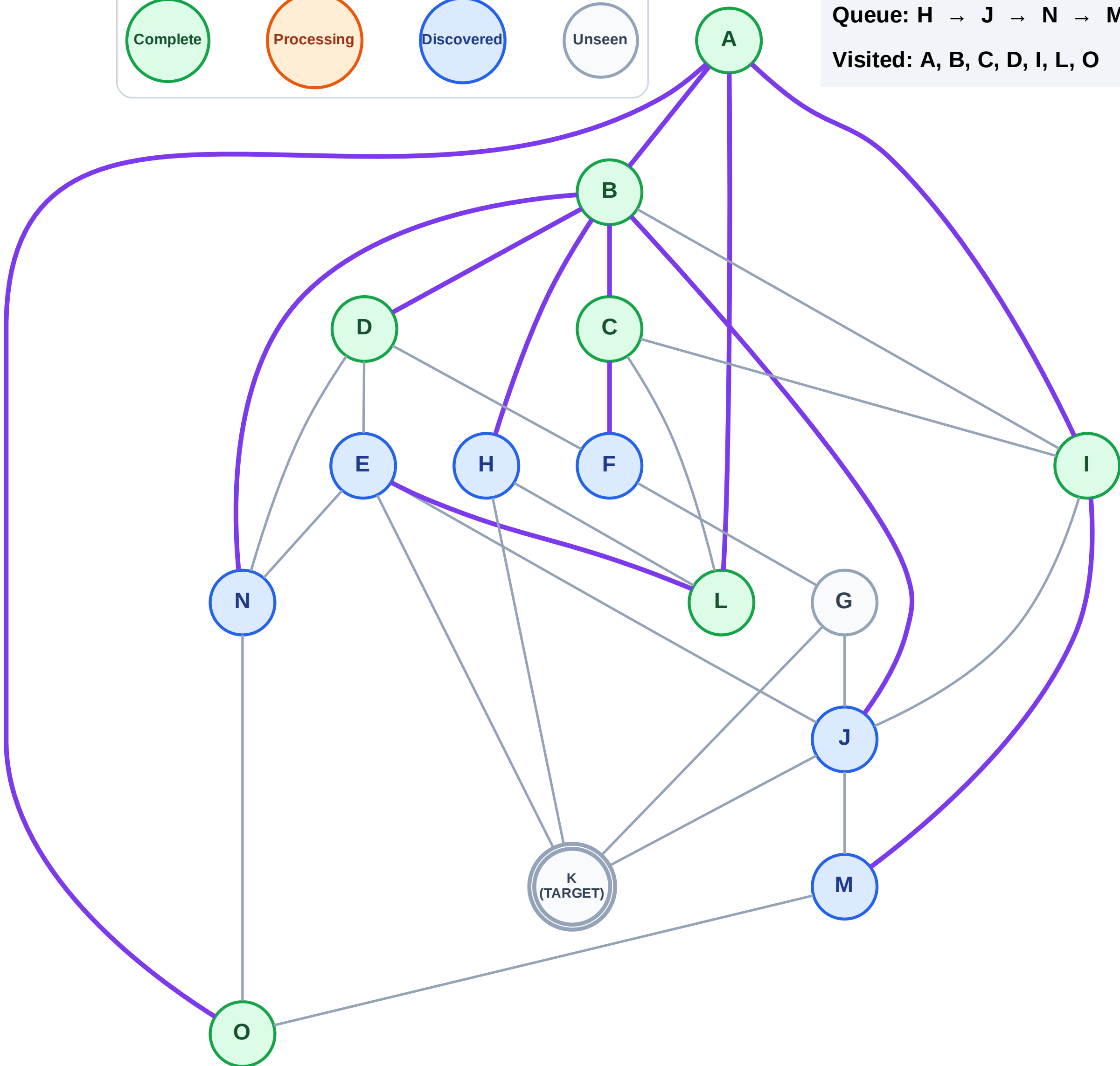
Step 15: No new neighbors from D

Legend



Queue: H → J → N → M → E → F

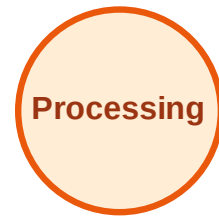
Visited: A, B, C, D, I, L, O



BFS Traversal

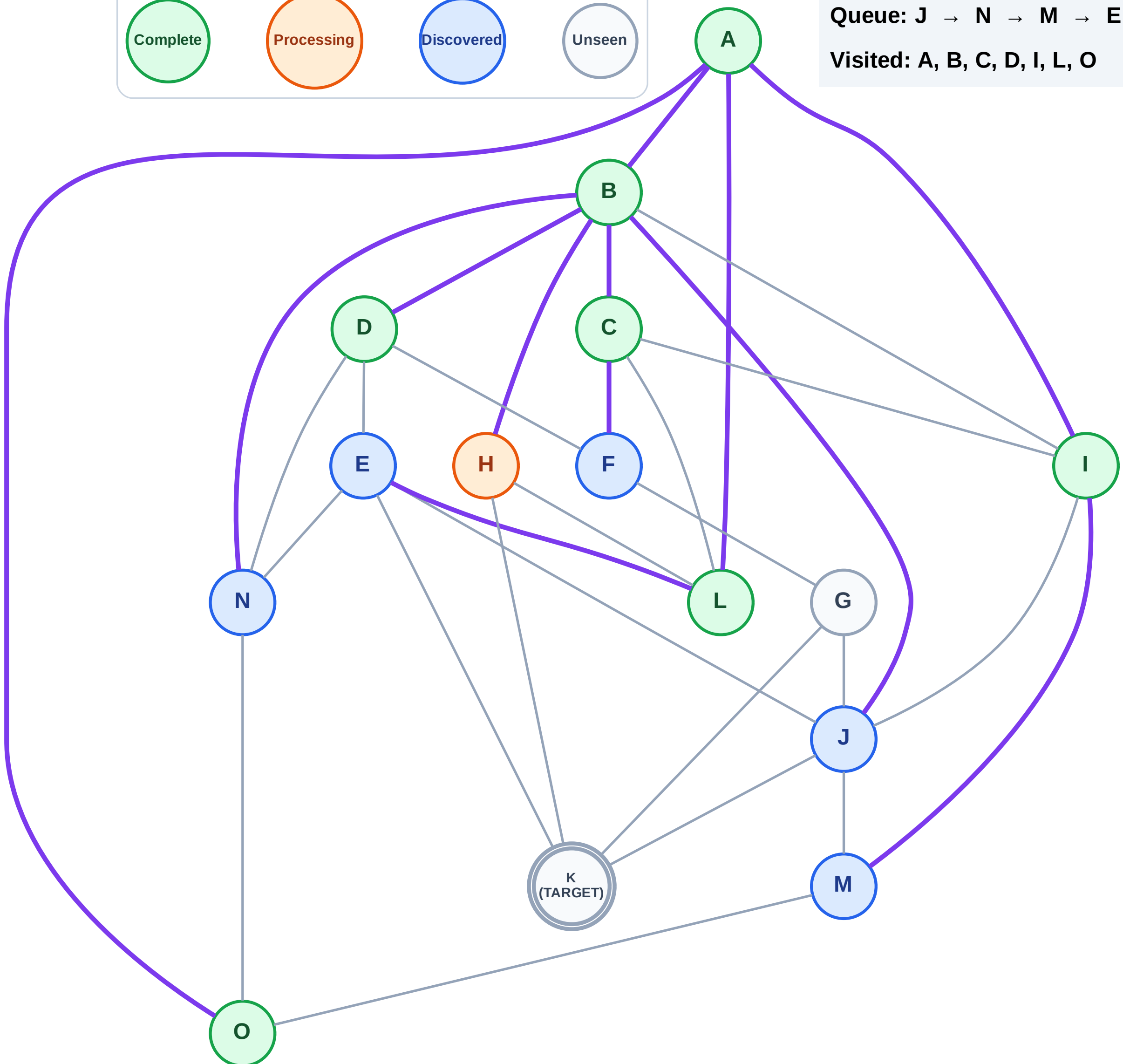
Step 16: Process node H

Legend



Queue: J → N → M → E → F

Visited: A, B, C, D, I, L, O



BFS Traversal

Step 17: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → H → K

Legend

Complete

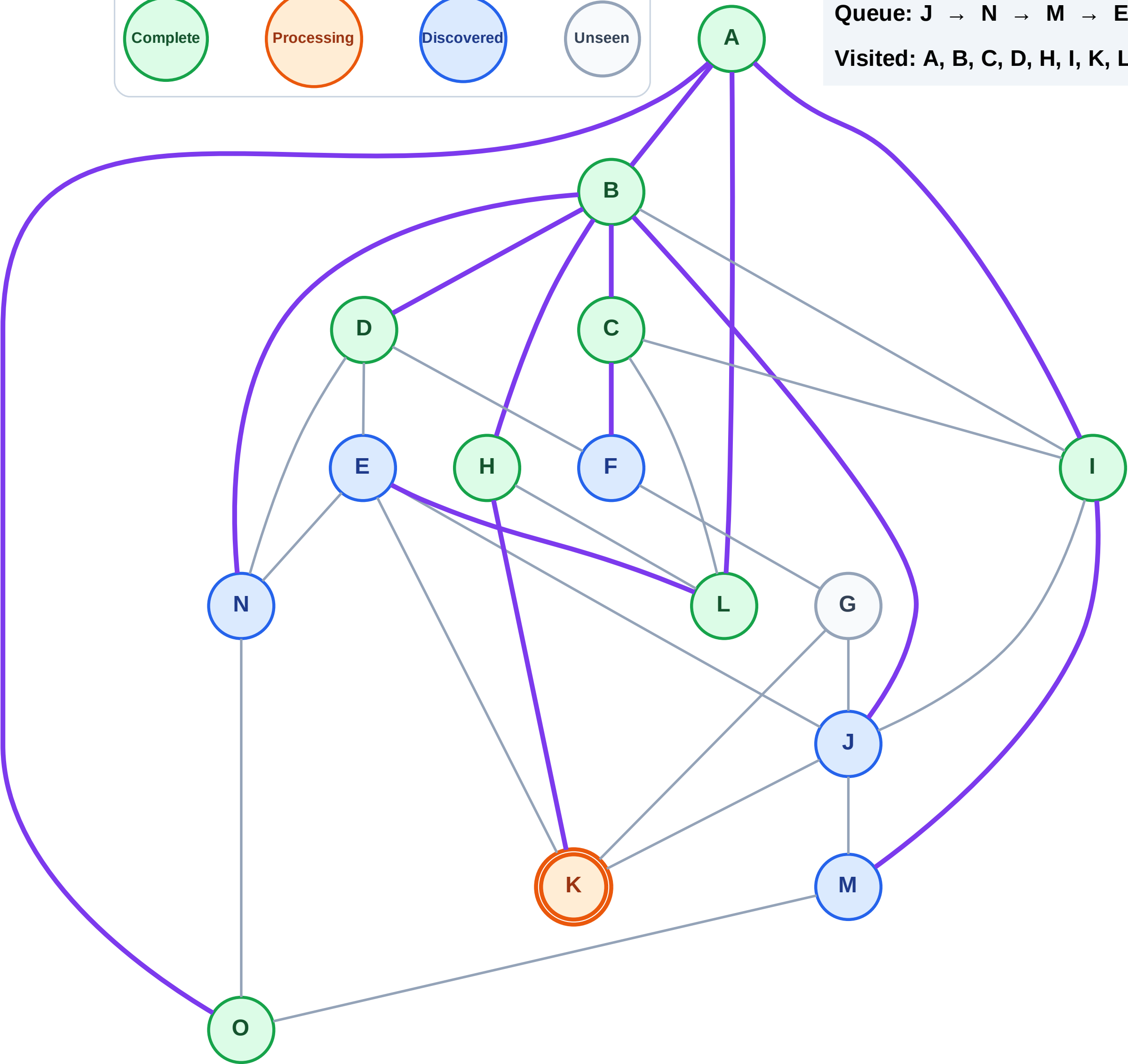
Processing

Discovered

Unseen

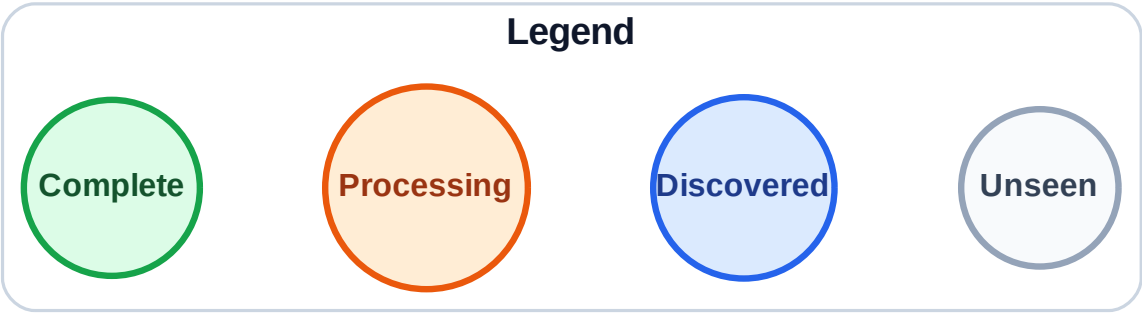
Queue: J → N → M → E → F

Visited: A, B, C, D, H, I, K, L, O

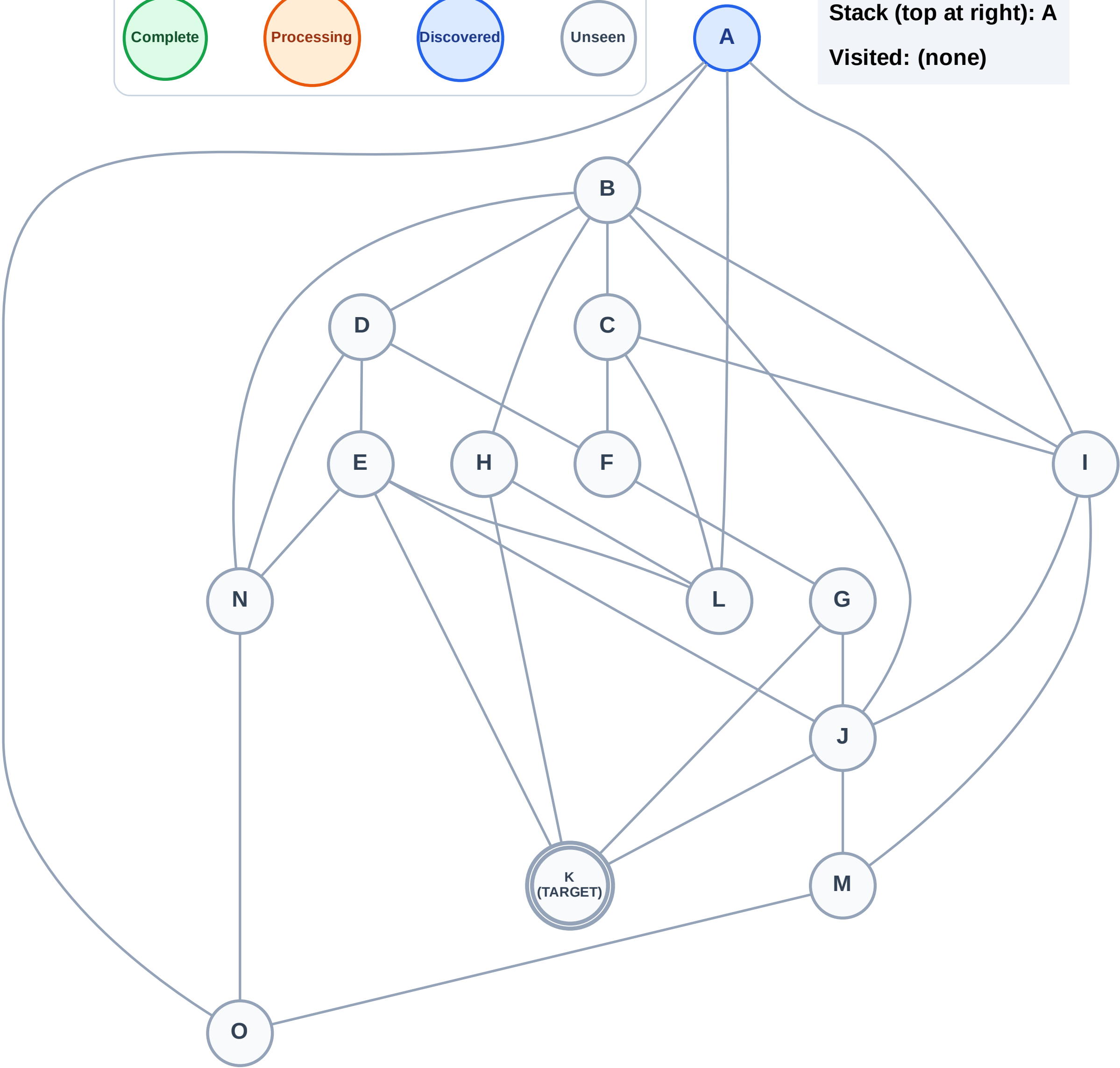


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

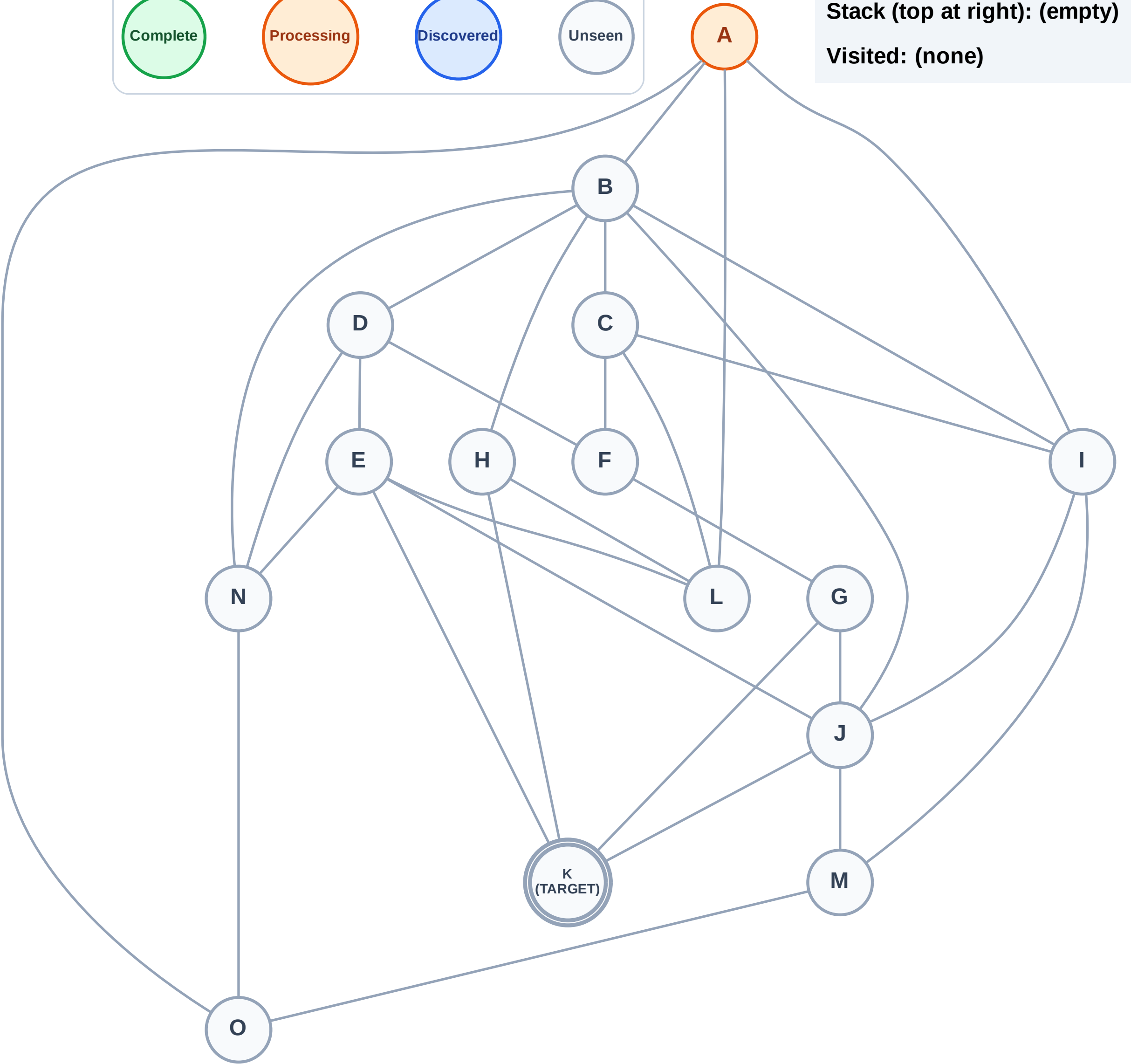
Complete

Processing

Discovered

Unseen

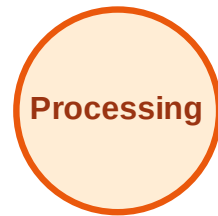
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

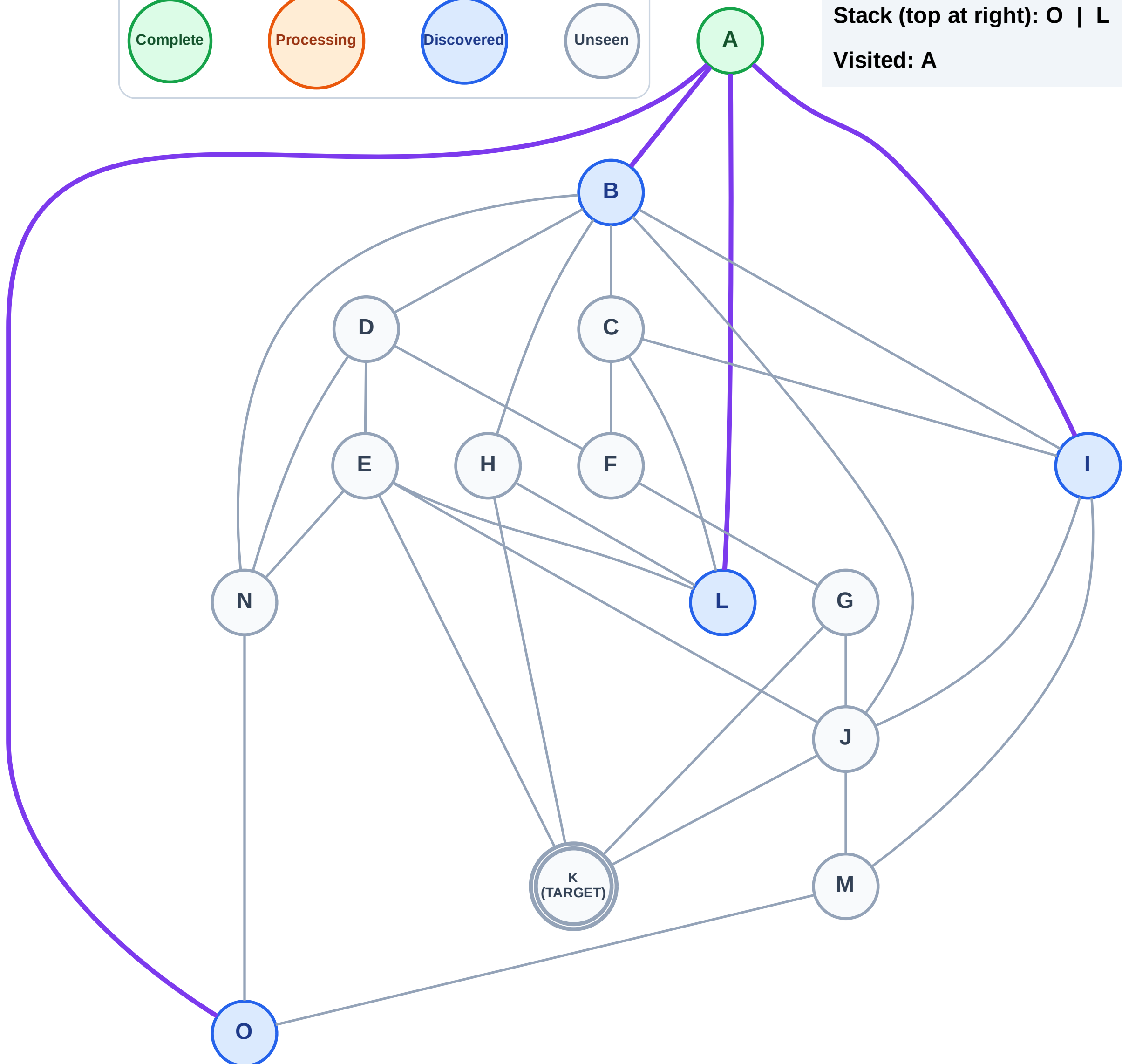
Step 3: Discover O, L, I, B from A

Legend



Stack (top at right): O | L | I | B

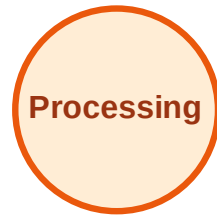
Visited: A



DFS Traversal

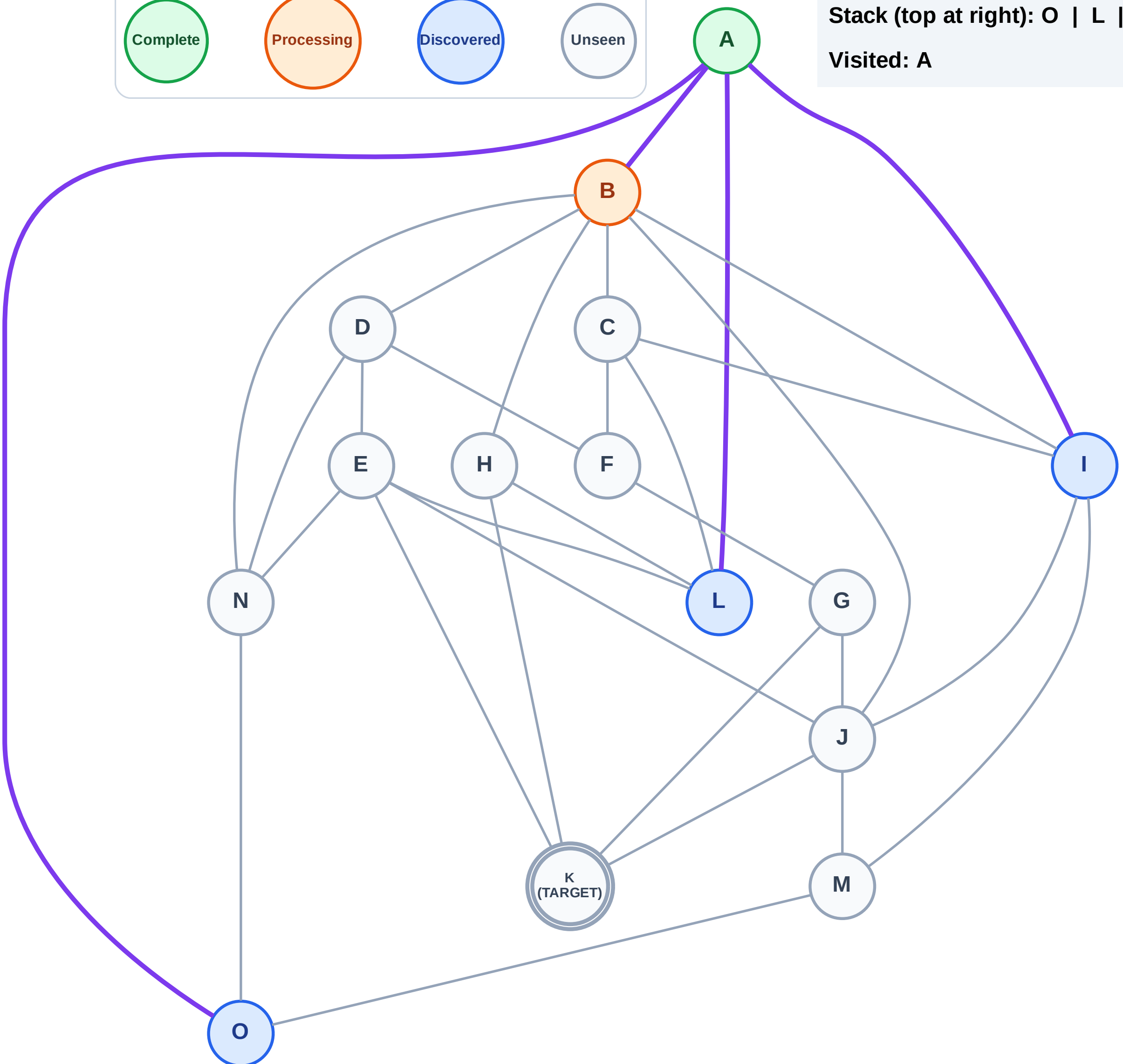
Step 4: Process node B

Legend



Stack (top at right): O | L | I

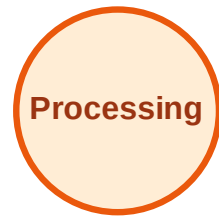
Visited: A



DFS Traversal

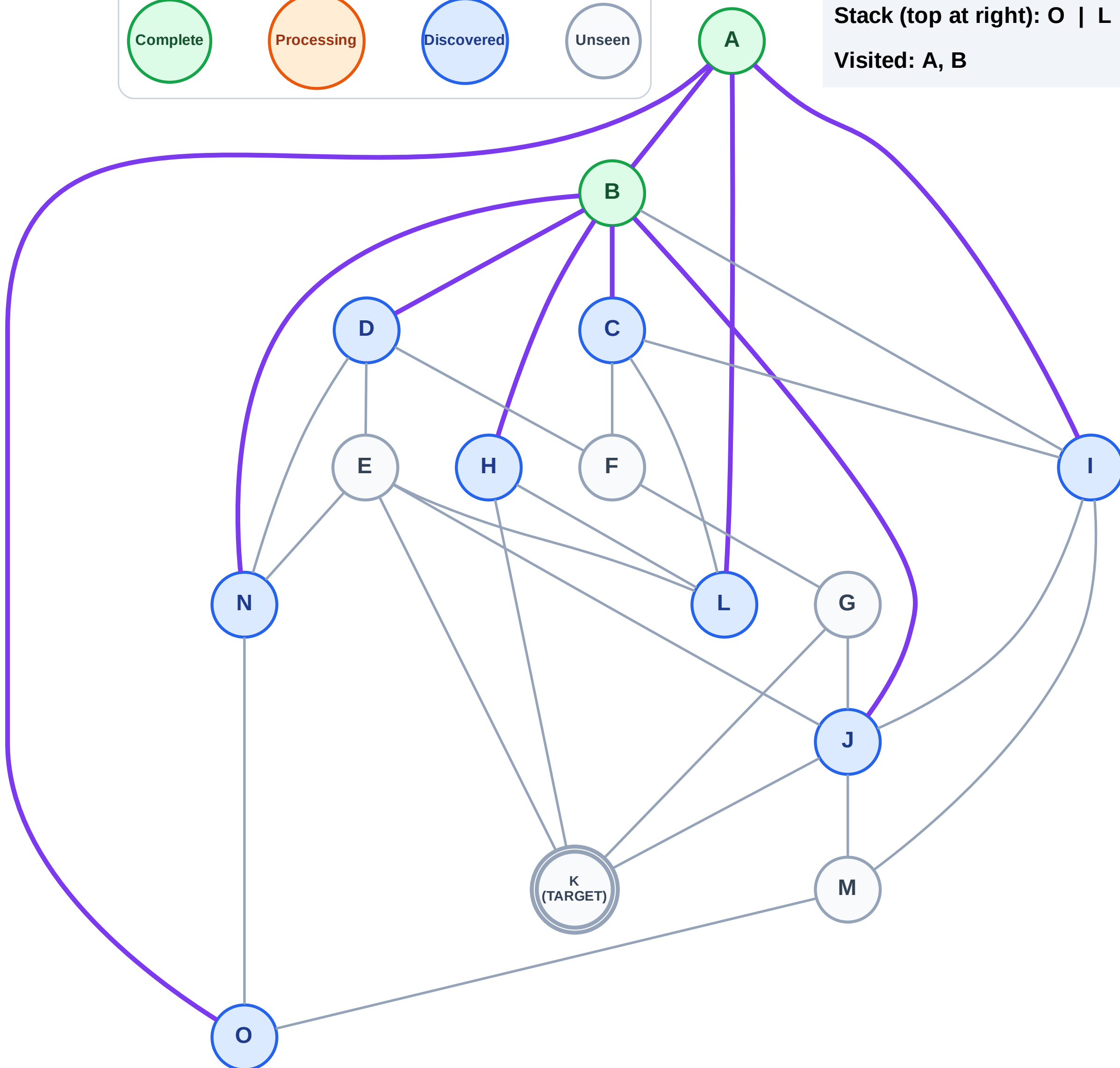
Step 5: Discover N, J, H, D, C from B

Legend



Stack (top at right): O | L | I | N | J | H | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

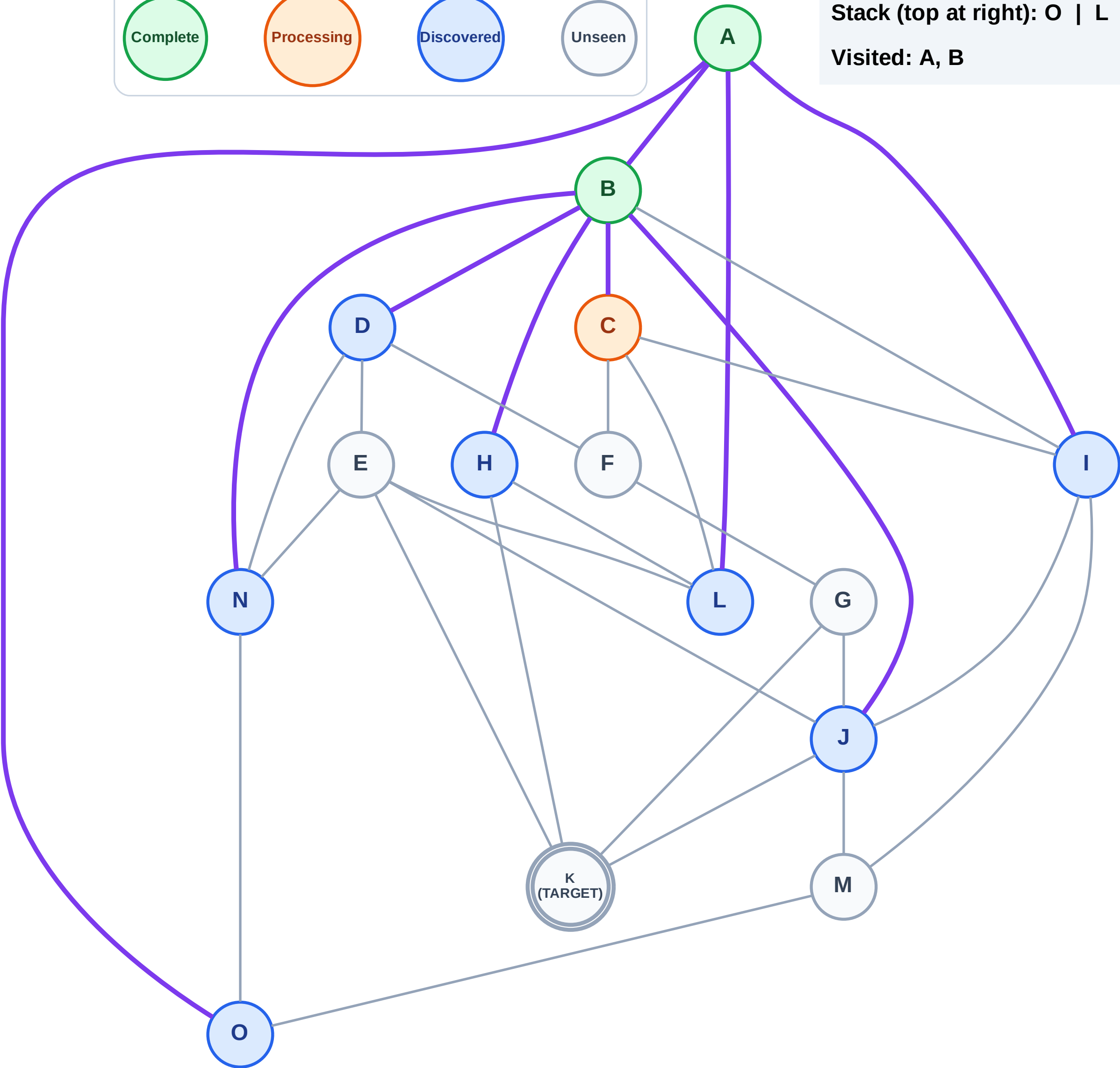
Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

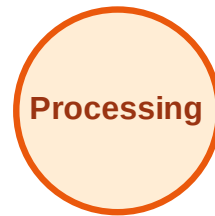
Visited: A, B



DFS Traversal

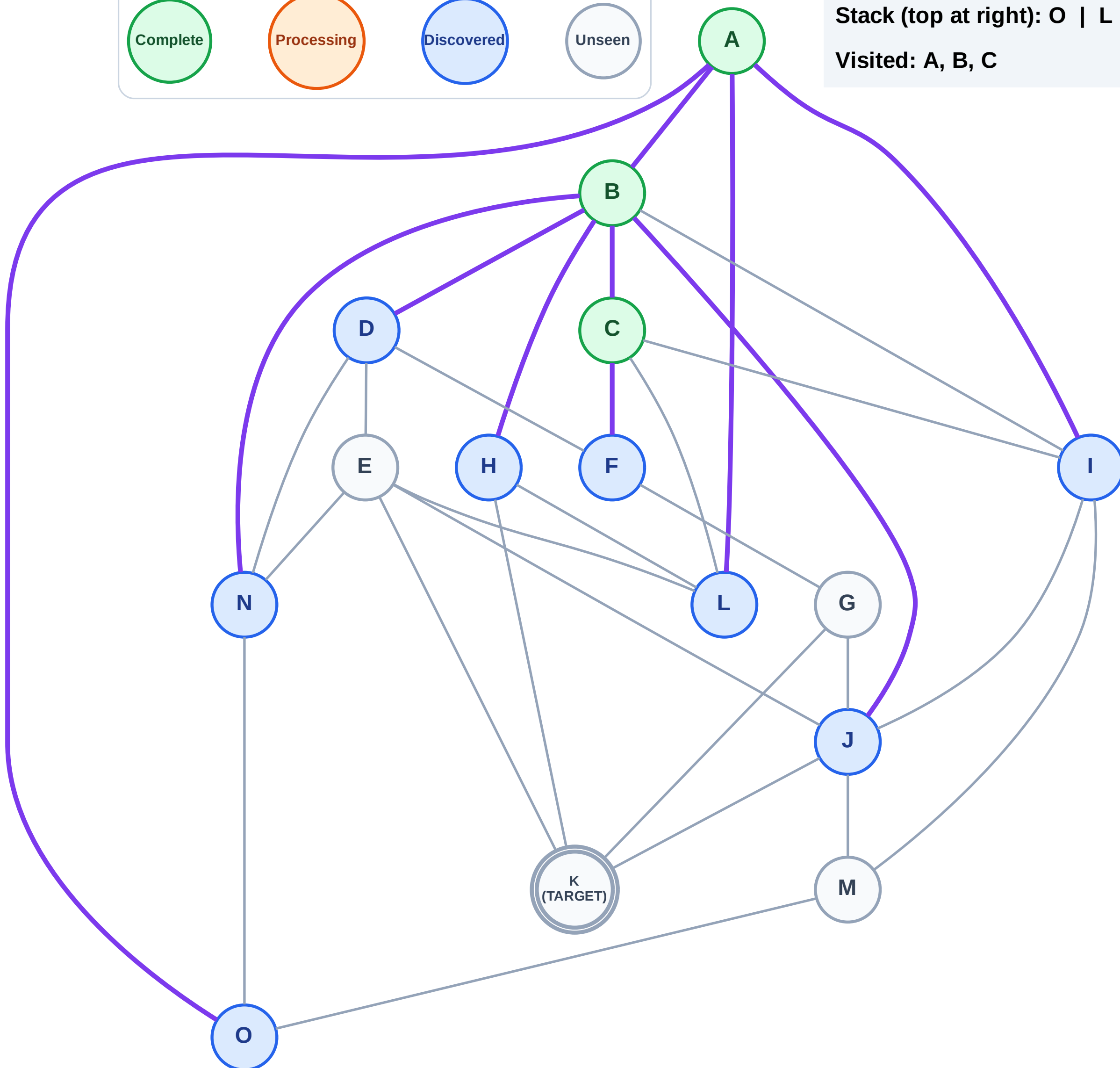
Step 7: Discover F from C

Legend



Stack (top at right): O | L | I | N | J | H | D | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

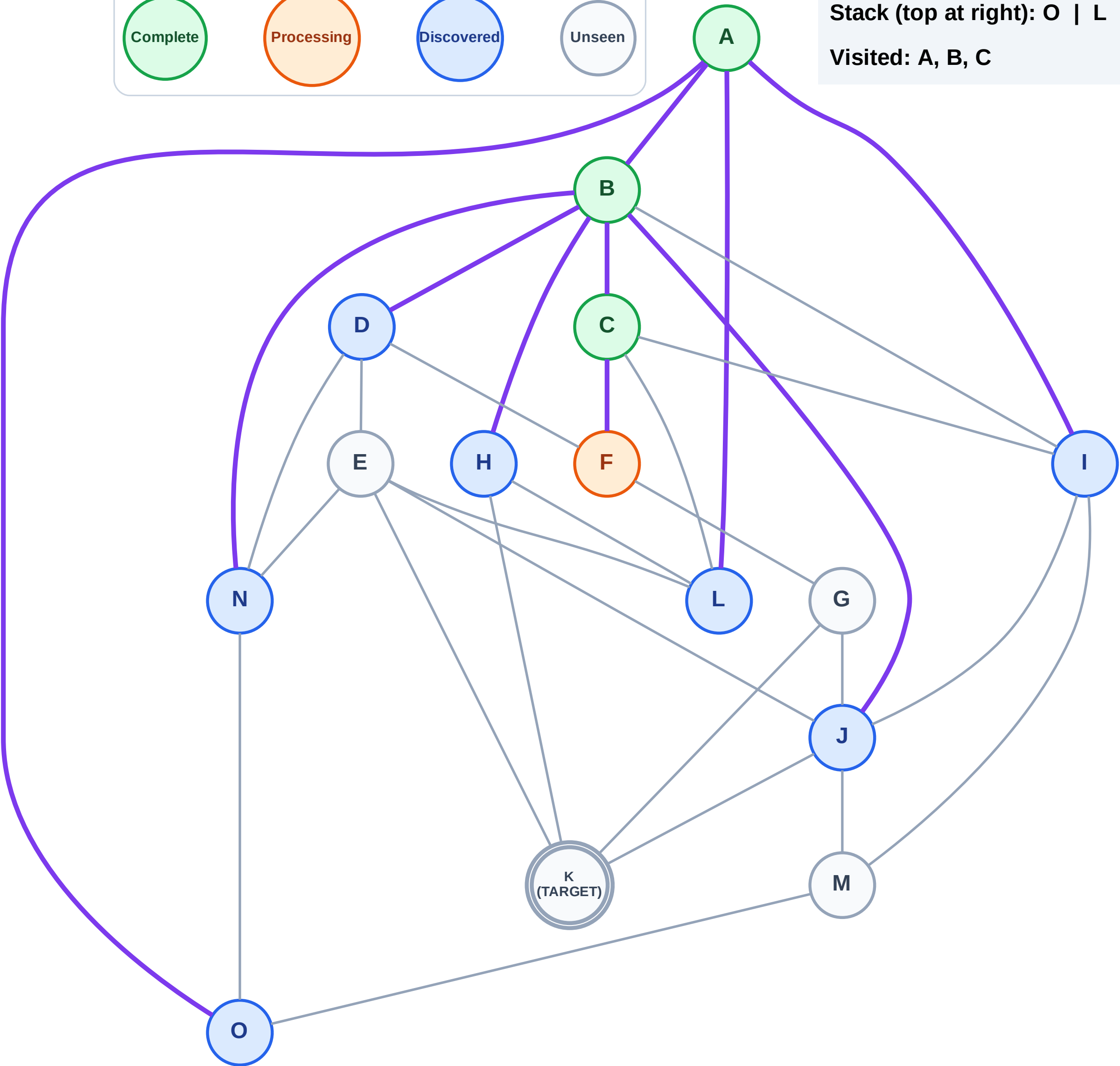
Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

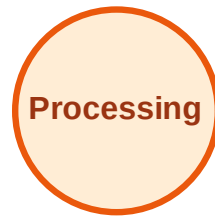
Visited: A, B, C



DFS Traversal

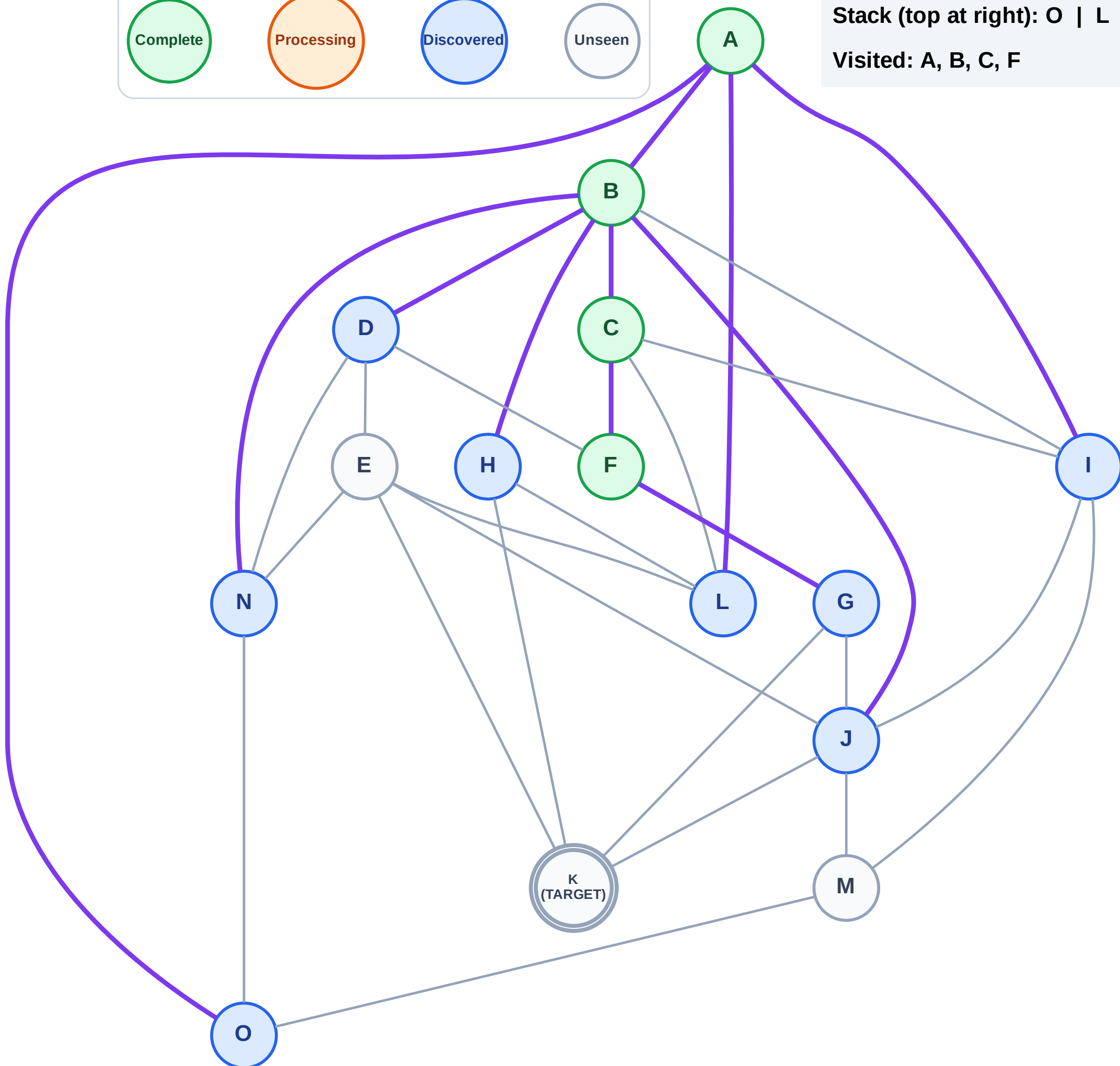
Step 9: Discover G from F

Legend



Stack (top at right): O | L | I | N | J | H | D | G

Visited: A, B, C, F



DFS Traversal

Step 10: Process node G

Legend

Complete

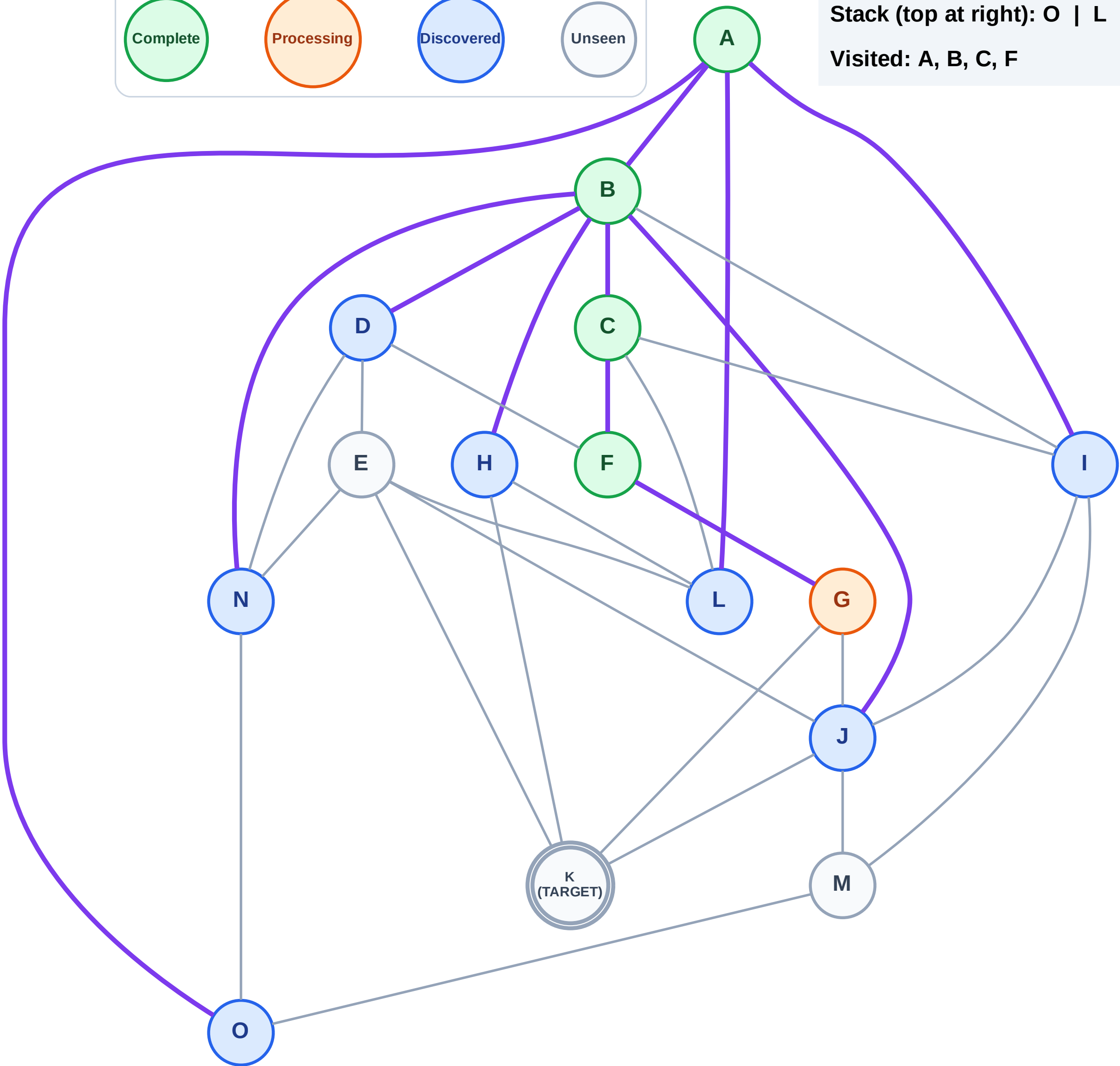
Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

Visited: A, B, C, F



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → C → F → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

Visited: A, B, C, F, G, K

